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Investigate Common Sources Of Noise During Image Acquisition, Analyze Their Effects On Image Quality

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Introduction to Image Acquisition Noise

Noise in image acquisition refers to unwanted variations that degrade the visual quality of an image.

Understanding the sources of noise is essential for improving image clarity and accuracy.

This presentation explores common noise sources and their impact on image quality.

NOISE AND IMAGES

The sources of noise in digital images arise during **image acquisition** (digitization) and **transmission**

- Imaging sensors can be affected by ambient conditions
- Interference can be added to an image during transmission



Electronic and Sensor Noise

Electronic noise originates from the inherent imperfections in the imaging sensor's electronic circuitry.

Sensor noise is influenced by temperature, sensor quality, and exposure settings.

This type of noise often appears as graininess or random speckles in images.

Environmental and External Sources

External factors like ambient lighting fluctuations can introduce variability in captured images.

Electromagnetic interference from nearby devices can cause signal disruptions.

These environmental noises can result in inconsistent brightness and color distortions.



Quantization and Compression Artifacts

Quantization noise occurs during analog-to-digital conversion, leading to loss of detail.

Lossy compression techniques can introduce blockiness and blurring artifacts.

Both types of artifacts diminish image sharpness and can obscure important details.

Effects of Noise on Image Quality

Noise reduces the overall clarity, making it difficult to interpret fine details.

It can obscure important features necessary for accurate analysis or diagnosis.

Excessive noise can lead to misinterpretation and reduce the reliability of image-based decisions.

Conclusion and Mitigation Strategies

Recognizing noise sources allows for targeted strategies to minimize their effects.

Techniques such as optimal sensor selection, proper lighting, and post-processing help reduce noise.

Continuous advancements in imaging technology are improving image quality despite inherent noise challenges.